

Integrated Credit & Market Risk Analytics Platform

Model Methodology - October 2026

1. Data

Credit risk: LendingClub Accepted Loans 2007-2018 (~2.26M loans). Filtered to terminal-status loans only (Fully Paid + Charged Off + Default) and 2014-2018 vintages, yielding 1,117,393 modeling observations. Time-based split: train 2014-2016 (891K), validate 2017 (169K), test 2018 (56K). Target = 1 if loan_status in {Charged Off, Default}, else 0.

Market risk: 9 ETFs (SPY, QQQ, IWM, EFA, TLT, HYG, GLD, USO, UUP) over 2014-2026 (3,118 daily observations) from yfinance. Extended history to 2007 for the 2008 GFC stress scenario.

2. PD models

Scorecard (logistic regression with WOE/IV)

Features binned via *optbinning.BinningProcess* with monotonic-trend enforcement and IV selection (min IV = 0.02). 12 of 21 candidate features survived. Logistic regression fit via *statsmodels.Logit* on WOE-transformed features (p-values required for adverse-action reporting). PDO scaling: base score 600 at 50:1 odds, PDO = 20 (industry standard).

XGBoost benchmark

XGBClassifier with native categorical support, early stopping on validation set. Hyperparameters: max_depth=6, learning_rate=0.05, subsample=0.85, min_child_weight=50, reg_lambda=1.0. Frozen train-time categorical vocabulary at inference to prevent unseen-value errors.

3. PD validation results (out-of-time, 2018 vintage)

Metric	Scorecard	XGBoost	Delta
AUC	0.6932	0.7130	+0.020
KS	0.2839	0.3139	+0.030
Gini	0.3864	0.4260	+0.040
Brier	0.1283	0.1258	-0.003

PSI train -> test = 0.030 (stable). Top decile predicted PD overshoots realized by ~14 percentage points, attributable to vintage maturity bias (2018 loans not fully matured).

4. LGD model

XGBoost regressor on realized_lgd = 1 - recoveries / funded_amnt for defaulted loans. Predictions clipped to [0.05, 0.95]. Out-of-time MAE = 0.07. R-squared near zero out-of-time, attributable to the structural difficulty of unsecured-credit LGD (recoveries are driven by post-default events not observable at origination).

5. Expected Loss and Basel IRB

$EL = PD * LGD * EAD$ per loan. Basel III IRB capital charge per dollar of EAD:

$$K = LGD * [\Phi(\Phi^{-1}(PD) + \sqrt{\rho} * \Phi^{-1}(0.999)) / \sqrt{1 - \rho}) - PD] * M_{adj}$$

Asset correlation ρ uses the Basel corporate formula. $RWA = K * 12.5 * EAD$.

6. Portfolio credit risk

Vasicek ASRF (analytical, Basel IRB foundation)

Each obligor's latent asset value $A_i = \sqrt{\rho} * M + \sqrt{1 - \rho} * \epsilon_i$; default if $A_i < \Phi^{-1}(PD)$. In the large-portfolio limit, conditional PD given an alpha-quantile factor draw has a closed form, which IS the IRB formula.

Copula Monte Carlo

Same one-factor structure as Vasicek, simulated. Gaussian copula uses $N(0,1)$ marginals; Student-t copula scales the Gaussian latent by $\sqrt{(\chi^2_{df} / df)}$, preserving joint tail dependence. Default threshold is $\Phi^{-1}(PD)$ or $T_{df}^{-1}(PD)$ respectively.

Validation

On a 1,000-obligor portfolio (PD 0.20, LGD 0.92, ρ 0.10, EAD \$15,000): Vasicek ASRF VaR(99.9%) = \$7.68M, Gaussian MC = \$7.81M (1.6% sampling noise). Student-t copula (df=5) VaR(99.9%) = \$9.20M, Economic Capital = \$6.44M, 28% above Gaussian under identical inputs. Headline 2008-modeling-story result.

7. Market risk (ported from standalone VaR project)

Five families: Historical Simulation, Variance-Covariance (Normal & Student-t), Monte Carlo (MV-Normal & MV-t), Filtered Historical Simulation with GARCH(1,1) residuals, and Extreme Value Theory via Peaks-Over-Threshold.

Out-of-sample backtest results (rolling 500-day window, GARCH refit every 60 days):

Method	99% exc rate	Kupiec p	Christof-ind p	Verdict
Historical	1.30%	0.142	0.001	Right rate, clustered
Parametric Normal	2.41%	0.000	0.000	Over-exceeds 2.4x
MC Normal	2.29%	0.000	0.000	Same
FHS GARCH(1,1)	1.13%	0.547	0.029	Passes both

EVT-GPD on 5% worst losses gives shape parameter $\xi = 0.31$ (heavy tail). EVT-extrapolated 99.9% VaR = 4.14%, more than 2x the Parametric-Normal estimate at the same confidence.

8. Known limitations

- 1) **Vintage maturity bias.** 2018 LendingClub loans not fully matured at the snapshot date; realized default rates are artificially low. The rigorous fix is a fixed observation window (e.g., default-within-24-months).
- 2) **LGD has limited signal.** Unsecured consumer LGD is dominated by post-default events; R-squared near zero on out-of-time test is structural. Mean is well-calibrated.
- 3) **Homogeneous portfolio assumption** in the /portfolio/credit_var endpoint. Heterogeneous portfolios are handled via the /credit/portfolio_el endpoint instead.
- 4) **Docker image ships with stub models.** Trained PD/LGD pickles must be generated locally; API gracefully falls back to calibrated stubs when pickles are absent.
- 5) **No counterparty / collateral framework.** Pure unsecured PD/LGD/EAD - real books need collateral haircuts and guarantor netting.